

Received 11 August 2025, accepted 24 August 2025, date of publication 29 August 2025, date of current version 8 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3603591

RESEARCH ARTICLE

Deployment and Performance Evaluation of 5G Private Networks, Enabling Use Cases in Rural Remote Areas

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This work was carried out within the COMECT project, which received funding from the European Union's Horizon Europe Research and Innovation Programme under Grant Agreement No 101060881.

ABSTRACT Rural remote areas have unique challenges for providing reliable and high-speed cellular connectivity, which makes them unattractive for mobile network operators. The Horizon Europe COMECT project has addressed connectivity in rural remote areas by 5G private network deployments, which so far have mainly been studied and deployed in industrial environment. The evaluation of the 5G private network deployments is conducted through field trials in four different environment, each of them characterized by different challenges and aiming to support different agro-forestry use cases. In particular, the evaluations relate to 1) indoor environments (e.g. hospitals and educational centers), 2) forestry, 3) livestock transportation, and 4) mines. The obtained results showcase that 5G private networks can support the proposed use cases. For example, in indoor environments, a downlink and uplink throughput of 700 Mbps and 50 Mbps, respectively, were measured, which are sufficient to enable e.g. remote monitoring of hospital patients. Moreover, it was measured that at least 10 Mbps in the uplink can be achieved for up to 300 m deep in the forest. The 10 Mbps target is also met in the farm location, including a satellite backhaul, which can enable e.g. the use case of (un)loading livestock from/to the truck. Additionally, throughput of more than 100 Mbps was measured at the mine location, which is sufficient to enhance safety by e.g. operating autonomous vehicles.

INDEX TERMS 5G private networks, field trials, forestry, indoor environments, livestock transportation, mines, network deployment, remote areas.

I. INTRODUCTION

In recent years, the fifth generation of mobile networks (5G) has seen wide deployment around the globe. Continuing the trend of the previous generations of cellular networks, 5G further improves broadband services as well as enabling a range of new services by also offering ultra-low latency and massive connectivity [1]. Typically, these so-called public

5G networks are deployed and managed by mobile network operators (MNOs), which then offer their supported services to individuals or businesses.

While MNOs provide a range of services, e.g. Internet access, and video streaming, through their 5G networks, their network deployments are mostly focused on urban areas. Providing connectivity in rural areas is unattractive and challenging for MNOs due to the high investment needed for the infrastructure deployment and the low population density in those areas [2]. Thus, rural areas often face limited or

The associate editor coordinating the review of this manuscript and approving it for publication was Nurul I. Sarkar¹.

poor connectivity, which in return impacts the residents and businesses, hampering economic development.

To allow more flexibility on the network deployment and the supported services of the 5G network, the concept of *5G private networks* or *5G non-public networks* (NPNs) has been introduced and standardized within the third generation partnership project (3GPP) [3], [4]. 5G private networks are localized networks, which are deployed within the premises of the private entity e.g. an organization or an enterprise, and they are customized to the needs of the private entity. Moreover, 5G private networks are managed by the private entity or a third-party provider, e.g. an MNO, on their behalf.

Due to the flexibility offered by 5G private networks, they have gained a lot of attention for deployment in industrial environments, which require enhanced security, and high communication availability [5], [6]. To that end, the 5G alliance for connected industries and automation (5G-ACIA) has identified different types of 5G private network deployments [7], and multiple deployment evaluations are performed e.g. in [8] and [9]. Even though there is high interest on 5G private networks for industrial environments, these networks are also *attractive to provide connectivity and new services in rural areas to support community development*. For example, different use cases can be enabled such as smart agriculture, rural remote healthcare, smart energy, environment monitoring, among many others [2].

A. RELATED WORK

Due to the recent conception and conceptualization of 5G private networks, several state-of-the-art works focus on simulations and theoretical models. For instance, Lim et al. [10] propose a deep reinforcement learning approach to optimize content allocation strategy and cache replacement policies in 5G private networks. Chang et al. [11] present an optimization algorithm to enhance the distribution and signal coverage of 5G base stations in mine deployments. Luo et al. [12] analyze the challenges associated with 5G private network design in the frequency range 2 band based on a game theoretic user allocation algorithm to minimize co-channel interference.

In parallel, practical work related to different fields such as drone scenarios and cyber security is emerging. For instance, Urama et al. [13] present the deployment of 5G private networks based on drone scenarios. Neinavaie et al. [14] exploit reference signals in private networks for cognitive navigation on ground and aerial platforms. Skokowski et al. [15] employ a 5G private network to test the robustness of the technology against low-energy and smart jamming attacks. Akgun et al. [16] study a machine learning algorithm for predicting cell interference by exploiting the channel state information reported by a 5G private network. Uitto and Heikkinen [17] evaluate video streaming in standalone 5G networks by measuring the latency and jitter under different video and network configurations. However their evaluation scenario is performed indoors and with a short distance

between the video equipment and the 5G modem, rather than in rural areas.

Despite previous work, very limited works are available on practical deployment of 5G private networks in rural areas. Among them, Schellenberger et al. [18] focus on an agricultural use case, where a drone and a robot are connected through a 5G nomadic network to a remote server. Even though the system architecture is validated as feasible, no detailed results on throughput and latency have been reported. To address the challenges and enable new applications in rural areas, Mendes et al. [2] propose the 5G-RANGE network, which is evaluated with field demonstration. The designed 5G-RANGE network proposes new physical, medium access control and network layers that are specifically designed for rural areas. Thus, 5G-RANGE can be used as a different operation mode for beyond 5G standards.

Differently, in our work we showcase how 5G private networks can be used to enable new applications in remote areas. In addition, several deployments are considered to provide an overview of the performance of these types of networks in rural areas. Section I-B describes in detail the main contributions and novelty of this study.

B. CONTRIBUTIONS AND PAPER ORGANIZATION

Within the COMECT project the investigations have been focused on evaluating the possible connectivity solutions, in this case 5G private networks, for different use cases in remote areas [19]. The approach taken was via field trials that could reflect, to the extend possible, the actual conditions in these remote areas. Previously, in [20], two use cases in remote areas, namely forestry and livestock transportation, have been described. For each of the two use cases, a network deployment has been proposed and evaluated with field trials. However, the evaluation was limited to the experienced uplink throughput. This paper is an extension of our previously published work [20] and the main contributions can be summarized as follows:

- 1) Description of *four* use cases in remote areas that can be enabled by the deployment of 5G private networks, and their associated network deployment. In addition to the two previously studied use cases, the deployment of 5G private networks in indoor environments, for e.g. rural educational purposes, and in mines is also studied in this paper.
- 2) Illustration of the performance and limitations of 5G private network deployments through field trials. The extended field test evaluation results include coverage analysis, uplink and downlink throughput, latency, and uplink throughput performance for the case of satellite backhauling link.
- 3) A cost benefit analysis of the deployment of 5G private networks.

In summary, this work is one of the first to report field trial-based evaluations across four different scenarios

TABLE 1. List of abbreviations.

Abbreviation	Explanation
3GPP	3rd Generation Partnership Project
5G-ACIA	5G Alliance for Connected Industries and Automation
AI	Artificial Intelligence
AMR	Autonomous Mobile Robot
GPS	Global Positioning System
(I)IoT	(Industrial) Internet of Things
KPI	Key Performance Indicator
MIMO	Multiple-Input Multiple-Output
MNO	Mobile Network Operator
(N-)LoS	(Non-)Line of Sight
NoW	Network on Wheels
NPN	Non-Public Network
PNI-NPN	Public Network Integrated Non-Public Network
RAN	Radio Access Network
RSRP	Reference Signal Received Power
SINR	Signal to Interference and Noise Ratio
SNPN	Standalone Non-Public Network
TCP	Transmission Control Protocol
TDD	Time Division Duplex
UDP	User Datagram Protocol
UE	User Equipment

for rural use case applications, offering valuable insights for both the academic research community and engineers at commercial operators. The key performance indicators (KPIs) and metrics measured offer valuable real-world data for model validation in simulation, cross-comparison in real environments, and insights into the deployment of private 5G networks in rural environments.

The remainder of this paper is organized as follows. In Section II the background of 5G private networks is presented including the major challenges from their deployment in remote areas. Section III presents the four use cases in remote areas that are under investigation and the associated network deployment. The performance evaluation of the proposed network deployments is presented in Section IV, whereas the relevant cost benefit analysis is presented in Section V. Finally, Section VI concludes the paper. For ease of reading, all abbreviations used across the sections are summarized in Table 1.

II. 5G PRIVATE NETWORKS IN REMOTE AREAS

This section provides background information on 5G private networks, its types of deployment, advantages, and challenges for deployment in rural remote areas.

A. BACKGROUND

5G private networks are intended for non-public use, meaning that their utilization is normally targeting specific services or use cases. Typically, 5G private networks allow access only to a limited set of devices (e.g. devices from employees and Internet of Things (IoT) devices of an enterprise) in a limited area (or number of areas). Additionally, due to data sensitivity and security reasons, a private 5G network can be organized such that both the communication data and/or operations and management data are kept within the enterprise premises.

One of the main advantages of deploying 5G private networks in remote areas is that it allows for *high performance connectivity*. Reliable and high-speed connectivity is challenging in remote areas. Introducing 5G private networks could be a game-changer with several notable benefits. Firstly, residents and businesses could enjoy faster download and upload speeds, facilitating more efficient access to online services, video streaming, and large file downloads. Secondly, latency performance can be improved that is crucial for real-time applications like video conferencing, tele-health services, and remote monitoring, ensuring seamless communication and rapid response times. Finally, the larger local capacity would be very helpful as it could accommodate a vast number of connected devices simultaneously, ranging from smartphones and tablets to IoT sensors and autonomous equipment, meeting the diverse connectivity needs of the remote area communities.

Another advantage is the *high communication customization*. 5G private networks are customized such that priority or high-quality is given to the limited set of services that are of crucial importance for the business or the enterprise. Another example is localization features (or in future sensing) that can potentially revolutionize precision farming, forestry, or remote indoor/mines practices. This feature combined with high-resolution video connectivity could also enable remote monitoring and teleworking. Furthermore, the 5G private network can be customized to seamlessly integrate IoT devices such as various sensors and actuators to satisfy the needs of the local area services and subscribers.

B. TYPES OF DEPLOYMENT

5G private networks (or NPNs) can be deployed in different ways, depending on which part(s) of the network are shared (or not) with a public land mobile network [7], [21]. We identify the following two categories of private network (or NPN) deployments:

Standalone non-public networks (SNPN): These private network deployments do not share any of their functions with the public network, which allows full control of the private network. SNPN deployments do not necessarily dictate that all network components are deployed at the enterprise premises. Different deployment flavours exist that can host some of the network functions outside the premises.

Public network integrated non-public network (PNI-NPN): These private network deployments share some of their functions with one or more public networks. Different flavours of PNI-NPN deployments exist, depending on which functions are shared with or hosted by the public network(s) and where these network functions are hosted. For example, in [7], the “shared radio access network (RAN)” deployment is presented, where the private network shares its RAN with the public network. Moreover, a special case of PNI-NPNs are the public-network-hosted 5G private networks, also known as “virtual 5G private networks”, where all network functions are embedded in the public network.

The two categories of 5G private network deployments are designed to accommodate different total cost of ownership and scaling/customization dependency on the MNO operating the public network. Consequently, some deployments have a low scaling/customization dependency on the public network, but a high total cost of ownership, with the extreme case of SNPNs having no dependency on the public network. On the other hand, the deployments that share many functions with the public network, have a low total cost of ownership but a high scaling/customization dependency on the public network. For example, the extreme case of virtual 5G private networks are fully dependent on the public network.

C. CHALLENGES

For the deployment of 5G private networks in remote areas, different challenges need to be addressed, for example, acquisition of spectrum licenses for SNPNs, coverage for PNI-NPN with shared RAN, bandwidth size selection, and backhauling for SNPN. Some of these challenges are addressed in this paper through the performance evaluation of different 5G private network deployments to support use cases in remote areas.

One of the challenges of 5G private networks is to identify the most appropriate deployment, i.e. SNPN or PNI-NPN, based on the requirements of the use case as well as the total cost of ownership, which includes both capital and operational expenditure. Typically, there is a trade-off between network performance and total cost of ownership. Therefore, a clear understanding of this trade-off is required per use case, as well as the definition of business models. This paper addresses this challenge by proving a cost benefit analysis in Section V.

Moreover, the performance of the deployed 5G private network, depends on a number of parameters. For example, the network coverage should be large enough such that all devices in the intended area can communicate with the network. Typically a location is considered as “covered” by the network if the reference signal received power (RSRP) at that location exceeds a threshold. Therefore, to guarantee sufficient coverage and capacity, the location of the base stations, and the coverage optimization of the sectorized antennas per allocated carrier frequency are essential. However, for PNI-NPN deployments, where the RAN (and thus, the base stations) of the public network is shared with the private network, sufficient coverage and capacity tailored to the localized “private” area of the particular enterprise could be challenging. This is due to the dependency on the public network’s RAN infrastructure that is potentially optimized to serving other “public” users outside the “private” area. On the contrary, with SNPNs, there is full control within the 5G private network in terms of base station location and antenna direction to achieve the required coverage and capacity. For instance, in dense environments with strong multipath components, such as indoor environments, signal blocking or non-line-of-sight (NLoS) can strongly affect the dynamic range and RSRP,

leading to poor coverage [22]. In this paper, coverage results are presented for each of the deployed networks, covering both SNPNs and PNI-NPNs.

Apart from coverage, the network performance also depends from the RAN configuration. In particular, in this paper, we address the challenges of identifying the most appropriate bandwidth size and time division duplex (TDD) frame configuration, which are directly related to the use case requirements. Both the bandwidth size and TDD frame configuration have a direct effect on the achievable uplink and downlink throughput, with the TDD frame configuration defining how many time slots are allocated to the uplink and downlink channels within a given time period. When deploying SNPNs, there are strict regulations in terms of 5G frequency license, which only allow for bandwidth sizes in a multiple of 10 MHz [23] and/or having a maximum allowed bandwidth size. Moreover, the license conditions include requirements in terms of TDD configuration, due to synchronization restrictions with other network operating in close proximity [24]. This could also restrict the performance of 5G private networks, especially for use cases where there is mostly uplink-oriented traffic. Both of this aspects are studied in this paper.

Last but not least, the availability of backhaul links is also critical, as they are required for the connection of the 5G private network to the Internet. This can be especially difficult in remote areas, where the terrestrial infrastructure is limited or non-existent. A potential solution for backhauling is the use of satellite communications. However, the performance of satellite communications may still be limited, regardless of the many recent technological advancements. Considering that the backhaul could be the limiting factor in the performance of 5G private networks, it has been chosen for study in this paper.

III. 5G PRIVATE NETWORK ENABLED USE CASES

This section describes the use cases, the challenges of each use case, and their associated 5G private network deployments. The methodology and setups presented allow for the reproducibility or comparison of deployments under similar conditions, enabling future evaluations in similar contexts.

A. INDOOR ENVIRONMENTS

1) USE CASES

The concept of wireless factories has spread significantly in the last decade. Due to advances in cellular connectivity, highly reliable automated production lines can now be implemented according to the concepts introduced by Industry 4.0 and the industrial Internet of things (IIoT) [25]. Among the connectivity solutions proposed to provide high reliability in wireless environments are 5G private networks due to dedicated spectrum and infrastructure [3].

Among the use cases, we highlight the massive connection of sensors with IoT devices for real-time monitoring,

allowing maintenance work and efficient resource management. Additionally, this connectivity can be extended to the control of autonomous mobile robots (AMRs) in charge of logistics in factories [26]. Finally, this highly reliable connectivity could enable augmented reality and digital twins for remote assistance and control in the production chain.

This first use case, focused on indoor environments, is presented as a baseline for the performance of a private 5G network compared to outdoor environments (see following use cases). Establishing this indoor baseline is essential for subsequent comparisons with deployments in more challenging rural or remote environments, where factors such as propagation, interference, and infrastructure limitations can significantly impact performance. From a high-level perspective, a 5G private indoor network can offer multiple use cases beyond those already explained in industrial environments, especially in rural areas where indoor environments like greenhouses or local healthcare and education centers can benefit greatly from advanced connectivity. Some examples are: (i) automation and sensor networks in agro-forestry applications (e.g., smart greenhouses requiring reliable low-latency connections for precise climate control or automated irrigation), (ii) connectivity in healthcare centers and hospitals for remote monitoring or remote surgery, which have critical ultra-low latency requirements, or (iii) connectivity in education centers where virtual and augmented reality classes are held, which may require high bandwidths [27], [28], [29].

Given the previous use cases, this paper analyzes an indoor 5G private network to assess the coverage and performance in terms of latency and throughput that such a deployment can provide in a realistic indoor scenario such as an industrial environment.

2) NETWORK DEPLOYMENT

The AAU 5G Smart Production Lab is a 1200 m² research laboratory located at Aalborg University (Denmark), which emulates the physical conditions of an industrial environment. Therefore, these facilities emulate an industrial setting with multiple production lines, robotic arms or autonomous mobile robots. Due to the research nature of the environment, there are multiple connectivity solutions deployed such as MULTEfire, Wi-Fi 6 or private 4G and 5G networks.

In the framework of this work, a 5G private network deployed in the lab has been assessed. This private network is connected to the public core of a Danish operator through a dedicated core-RAN over a fiber connection. This solution, deployed in the 3.7 GHz in frequency range 1 n78 band, has a bandwidth of 100 MHz, a TDD frame configuration of 3 uplink and 7 downlink slots, and a subcarrier spacing configuration of 30 kHz. The private network hardware deployment consists of a Nokia Mxie 5G SA core and a Nokia AirScale indoor radio unit, which can be visualized in Figs. 1(a) and 1(b), respectively. The indoor radio unit, which acts as an access point, provides coverage to one of the halls

of the facility with an approximate area of 400 m², and is located at a height of 5 m.

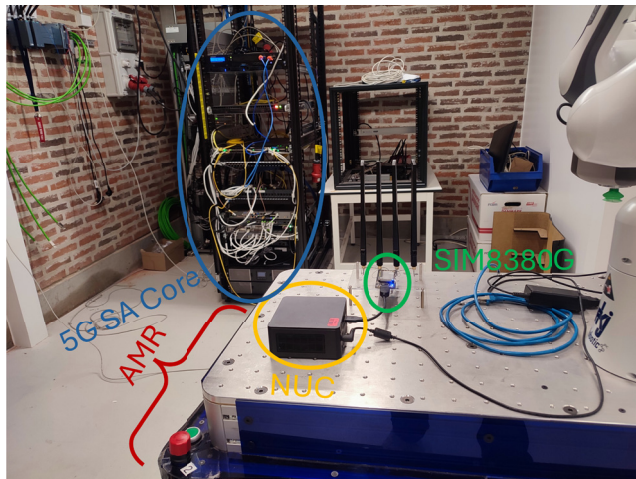
To analyze the performance of the described private network, a SIMCOM SIM8380G-M2 modem with support for 5G NSA/SA in the network operating band is used as user equipment (UE). Four antennas are connected to this UE to take advantage of the modem's multiple-input multiple-output (MIMO) capabilities, as illustrated in Fig. 1(a). To monitor UE operations, the UE is connected to a small computing box (next unit of computing) from which *ping* and *iperf* commands run to evaluate network performance. Both the modem and the computing box are placed on an AMR with the ability to navigate the 5G SmartLab following predefined routes due to a LiDAR-based positioning system (see Fig. 1(a)). While the AMR stores data related to its positioning within the laboratory, the modem stores data related to network performance and signal quality. Therefore, by synchronizing the timestamps from both ends it is possible to accurately map the network performance parameters at each location in the industrial scenario. The analysis of the radio propagation characteristics and performance of the 5G private indoor network is presented throughout Section IV.

B. FORESTRY

1) USE CASES

In Norway, forests cover roughly 33% of the country's land area, making them a crucial part of both the economy and environmental management efforts. Sustainable forest management is central to Norway's strategy, supporting the long-term vitality of the forests while enabling a wide range of economic activities. However, the forestry sector in Norway faces a significant challenge in terms of digitalization. The sector's value chain has not yet fully embraced technological innovations, particularly with regard to connectivity solutions [30]. As Norway continues to foster innovation across various sectors, there is an opportunity to boost the forestry industry's resilience and sustainability by adopting digital transformation and keeping pace with technological developments. The COMTECT project has identified two key use cases where 5G private networks could play a pivotal role, as depicted in Fig. 2.

Remote Operational Support for Forest Machinery Operators: The forestry industry faces a critical need for remote expert assistance during thinning and logging activities, where operators in the field require immediate guidance and support. This challenge is compounded by the limited coverage and performance of public mobile networks in forested regions. To address this, the proposed solution involves deploying a mobile 5G private network on wheels (NoW), specifically designed for forestry environments, as shown in Fig. 2. This 5G private NoW facilitates high-quality video transmission from forestry machinery, enabling real-time remote assistance through online cameras. Experts from the forest operator's company can guide on-site operators in making informed decisions. Additionally, this solution



(a)



(b)

FIGURE 1. (a) Photograph of the SIMCOM modem and the next unit of computing (NUC) assembled on the AMR, and the 5G SA core located in the 5G SmartLab, and (b) scenario through which the AMR navigates. AirScale indoor radio (ASiR) is illustrated on the roof of the factory.

facilitates remote maintenance, diagnostics, and repairs of machinery, with vendor experts providing supervision via high-definition video feeds.

Enhanced Situational Awareness in Forests: The increasing risk of forest fires, exacerbated by climate change, underscores the need for a sophisticated monitoring and surveillance system in forested areas. Forests are vulnerable to a range of emergencies, including wildfires, landslides, and floods, necessitating advanced solutions to ensure the safety of both the environment and emergency responders, including police, ambulances, fire departments, and volunteers. Current systems require rapid reporting of incidents, such as fires, through digital channels. In response, the NoW solution shown in Fig. 2, integrated with drone technology and on-the-ground sensors, offers a comprehensive monitoring and surveillance platform. This system enables the swift notification of emergency personnel during critical incidents.

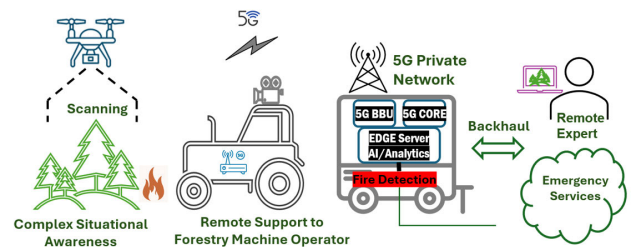


FIGURE 2. Depiction of forestry use cases.

High-resolution, multi-spectral cameras mounted on drones capture aerial imagery, which is transmitted in real-time via the private 5G NoW to an external edge server. The inclusion of on-map analytics further enhances the ability to visualize forest conditions, ensuring timely and effective responses to any alarming changes.

2) NETWORK DEPLOYMENT

The COMECT forestry use cases require substantial uplink throughput to support the transmission of high-quality video streams from remote forestry machinery and drones. To meet this demand, the concept of a 5G private network, specifically the NoW, has been introduced. The NoW functions as a self-contained system, integrating 5G radio, 5G core, and related applications within a single, highly mobile platform, as depicted in Fig. 3(a). This configuration enables the rapid deployment of a standalone 5G private network that delivers seamless 5G coverage. The traffic generated by NoW is backhauled through either terrestrial networks or satellite links, depending on the operational requirements and location, allowing remote access to cloud services or the internet for expert consultations. To enhance real-time forest monitoring and situational awareness, the NoW's Edge server can be equipped with artificial intelligence (AI) and machine learning algorithms to detect and respond to incidents such as forest fires or other anomalies efficiently.

To evaluate the suitability of NoW for forestry applications, multiple tests were performed with the NoW, providing both 5G radio and 5G core functionalities, deployed in a selected forest location. The tests were conducted using 80 MHz of bandwidth in the 3.3–3.4 GHz frequency range, with a transmit power of 35 dBm and an antenna height of 2.5 m. An uplink-heavy TDD frame configuration of 3 uplink and 7 downlink slots was utilized to prioritize uplink performance. The measurement locations within the forest are shown in Fig. 3(b). Due to the dense forest canopy, measurements could not be taken farther from the NoW deployment point. The measurements were conducted using a Huawei P40 smartphone, acting as a UE, held at a typical natural height of 1.5 m. The tests employed the web-based OpenSpeedTest tool,¹ which was hosted on the NoW's Edge server. This tool measures uplink and downlink throughput,

¹<https://openspeedtest.com/selfhosted-speedtest>

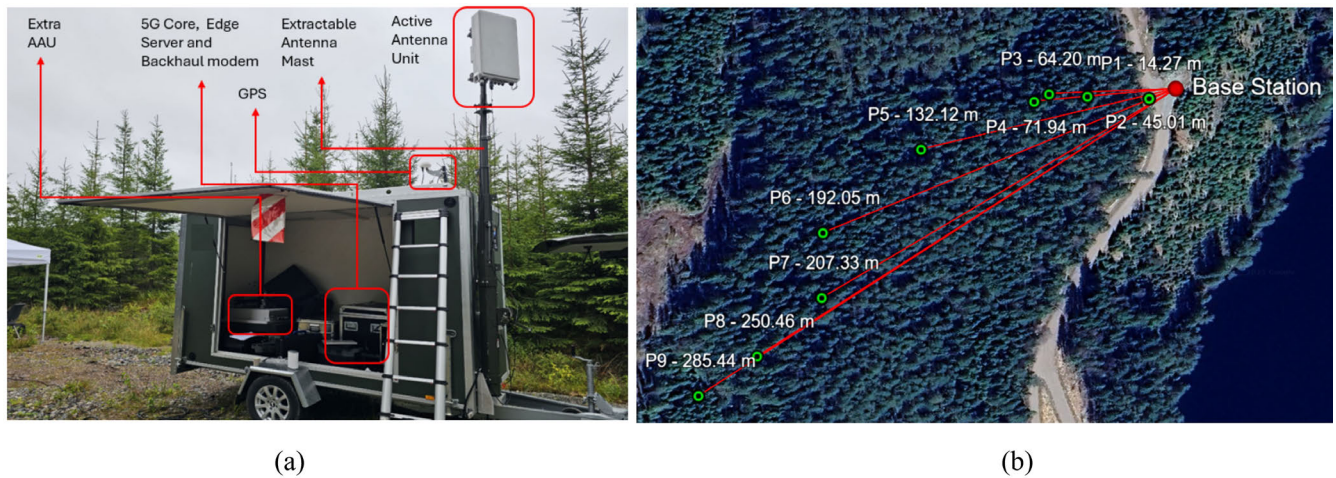


FIGURE 3. (a) Network on Wheels (NoW), and (b) measurement points in a dense forest with high vegetation, where the base station marks the location of the NoW.

and latency, directly through a browser, without the need for additional software. These measurements are performed using multiple transmission control protocol (TCP) streams. The distance between the NoW and the UE was calculated using global positioning system (GPS) coordinates, with an accuracy of approximately 3 m. Notably, no backhaul was involved in this phase of testing, as the focus was on evaluating the performance of the 5G private network itself. The performance evaluation is presented in Section IV.

C. LIVESTOCK TRANSPORTATION

1) USE CASES

One of the major industries in Denmark is the pig industry, which is highly regulated by both the European Union and the Danish authorities to ensure the welfare of animals before, during and after their transport [31], [32]. Within the COMECT project, discussions have been taken place with stakeholders from the livestock transportation industry and two potential use cases have been identified that will be beneficial to the industry.

Automatic license plate recognition: For a transportation truck to be allowed to enter a piglet (un)loading location, a check on the license plate number should be performed. This process is important as trucks need to go through a disinfection process before being allowed at a (un)loading location and the truck's clearance is registered to a database, coupled to their license plate number. As a solution, it is envisioned that a camera will be installed on and control the road barrier leading to the (un)loading location. The camera will upload the image of the license plate to a remote server, which is connected to the database and where the relevant software runs.

Automatic animal counting and monitoring: The animals are being counted during their (un)loading from/to the truck, and thus an efficient and correct animal counting is important. Moreover, during the (un)loading process, it is

beneficial to monitor and detect the status of the animals, e.g. injuries, to ensure that their health status remained unchanged during their transport. The envisioned solution includes the deployment of a camera at the (un)loading location, which will live stream a video of the (un)loading process to a remote server, where the relevant software will run.

The envisioned deployment solutions for both use cases is similar i.e. deployment of a camera that will upload an image/video to a remote server. From a communications point of view, sufficient uplink throughput, and thus coverage, should be ensured both at the road barrier and at the (un)loading location. In terms of uplink throughput, the 10 Mbps [33] have been identified as a minimum requirement, whereas for coverage, typically, an RSRP target of -120 dBm is chosen.

2) NETWORK DEPLOYMENT

To support the livestock transportation use cases, the connectivity solution in Fig. 4 is proposed. Specifically, local connectivity at the (un)loading location is provided with a 5G private network, which is then connected to the remote server via terrestrial or satellite Internet access. The choice of backhaul depends on the availability of terrestrial networks at the (un)loading location, which is typically in rural areas with little or no cellular coverage. In this work, and in relation to the livestock transportation use case, we assess the performance of a 5G private network in terms of coverage and uplink throughput. Moreover, we assess the performance of the backhaul via satellites, as the backhaul can significantly limit the experienced end-to-end uplink throughput.

For our evaluations, we performed field tests at a farm location in The Netherlands, where the Amarisoft CallBox Classic,² illustrated in Fig. 5(a), was used to act as an SNPN,

²<https://www.amarisoft.com/test-and-measurement/device-testing/device-products/amari-callbox-classic>

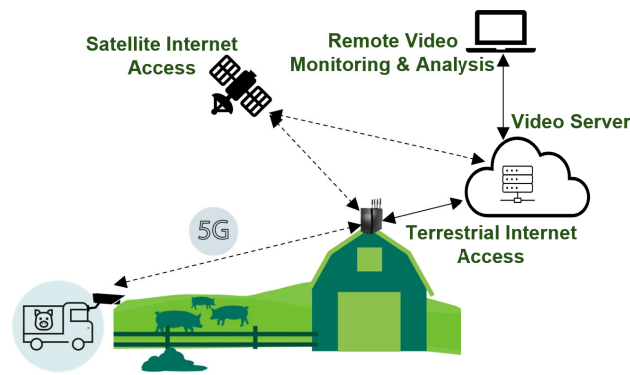


FIGURE 4. Livestock transportation use case with a 5G private network.

providing both 5G radio and 5G core functionalities. The Amarisoft was configured in the n77 band, in particular between 3.825 GHz and 3.875 GHz and its maximum transmission power was -20 dBm/MHz. For the measurements two different bandwidth sizes were considered, i.e. 20 MHz and 50 MHz. Moreover, two different TDD frame configurations were considered. The first TDD configuration is typically recommended for mostly downlink oriented traffic and consists of 3 downlink slots, 1 special slot and 1 uplink slot [24]. The second TDD configuration is an uplink heavy configuration as it consists of 5 downlink slots, 2 special slot and 3 uplink slots during a 10 slots frame [24].

To perform the relevant coverage and uplink throughput measurements, the Quectel RM502Q-GL has been used to act as a UE at different locations at the farm, which are illustrated in Fig. 5(b). For the assessment of the backhaul link, in terms of uplink throughput, the Starlink Standard V3 terminal³ has been connected to the Amarisoft CallBox to provide connectivity between the UE and a server located in Amsterdam. For all uplink throughput measurements, the iPerf⁴ has been used in user datagram protocol (UDP) mode, which is a commonly used protocol for video streaming codecs due to its latency characteristics and packet loss tolerance. The results on coverage, uplink throughput, and backhaul are presented in Section IV.

D. MINES

1) USE CASES

Open-pit or opencast mine extracts minerals from the surface instead of underground. Open-pit mining is the most common method used in Türkiye for mineral mining and does not require extractive methods or tunnels. This surface mining technique is used when minerals or deposits are relatively close to the surface of the earth. Open pits are sometimes called quarries when building materials and dimension stones are produced.

Open-pit mines are dug on different benches depending on the excavation machinery. The walls of the open-pit

mines are dug at an angle and include steps to prevent avalanches from occurring inside the building site. These walls in the open-pit mines are dangerous for the operators of the excavators and drivers of the heavy dump trucks. The goal is to decrease accidents and mistakes in the risky environment of Türkiye's open-pit mines with autonomous collaborative vehicles connected to a private network.

The private network solution is developed for automated and teleoperated vehicles in open-pit mines located in extremely suburban areas. With the support of 5G infrastructure, AI-supported central task planning, and smart sensors, autonomous collaborative vehicles will be developed and operated for excavation works.

2) NETWORK DEPLOYMENT

With the scalability of the 5G open-pit mine private network architecture, two experience zone activations have been carried out at high-profile events, where participants were able to experience the control of excavator from a remote location 800 km away, with a latency of one-tenth of the blink of a human eye. In addition to enabling 5G standalone edge network in the open-pit mine private network, the quality of services is guaranteed by AI-based resource optimisation of network slicing technology. The open-pit mine 5G private network solution is covered by five road side units and two fibre base stations with advanced cellular vehicle-to-everything use cases that AI on the edge applications and 5G standalone edge network scalable architecture. The open-pit mine private network deployment uses 100 MHz of bandwidth in the 3.5–3.6 GHz frequency range, with a transmit power of 55 dBm and an antenna height of 18 m. The deployment of five road site units is shown in Fig. 6.

The proposed private network aims to enable 5G connectivity along with an edge network architecture that provides AI-assisted perceived zero latency and ultra-reliability in open-pit mining areas. The private network deployment includes 5G radio equipment mounted on the base stations in the demo location, the transformation of 5G sites into 5G fibre sites, and conducting fault tests. AI-assisted perceived zero latency and ultra-reliability algorithm was developed with the data collection platform designed and located at open-pit mine private network. This platform was also designed to support custom data generation. The performance evaluation of the proposed private network is presented in Section IV.

IV. PERFORMANCE EVALUATION

This section presents the performance evaluation of the proposed 5G private networks to support the previously described use cases. The results are presented per KPI rather than per use case to more easily showcase the difference in performance between the different network deployments on a specific KPI. Moreover, network deployments have only been evaluated for the KPIs that are relevant for the associated use cases. Therefore, not all network deployments are compared for each KPI. The KPIs that are under investigation are coverage (in terms of RSRP measurements),

³<https://www.starlink.com/specifications>

⁴<https://iperf.fr/>



FIGURE 5. (a) Measurement setup with Amarisoft CallBox Classic and (b) measurement locations at the farm.

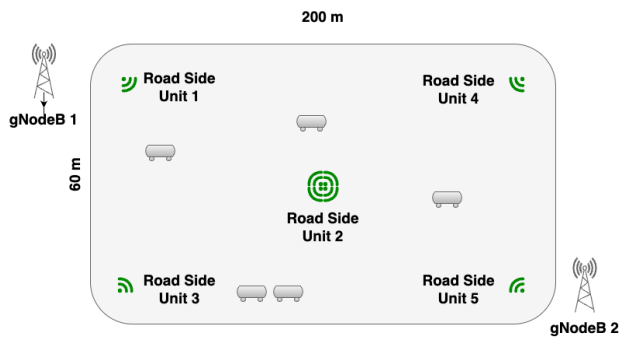


FIGURE 6. Network deployment of open-pit mine private network.

uplink throughput, downlink throughput, latency, and uplink throughput for a satellite backhaul.

A. COVERAGE

Coverage is an important KPI for all network deployments as it defines the area where the UEs can connect to the network. Typically, to determine the coverage area, RSRP measurements are performed at different locations, based on a reference signal transmitted by the network. The RSRP, and hence the coverage, depend among others on the transmit power, antenna gains, and the propagation conditions in the environment. Therefore, for each network deployment, the RSRP values over different distances are presented. Note that typically a location is considered as covered when the RSRP exceeds -120 dBm.

1) INDOOR ENVIRONMENTS

Given the setup explained in Section III-A and illustrated in Fig. 1, we evaluated the indoor coverage as the strength signal received along the hall in the industrial scenario. For this purpose, we proceed to navigate a route with the AMR in which the UE measures the RSRP level at different locations in the hall. Fig. 7(a) shows the range between the UE–access point link given the different AMR locations

within the lab. This range varies between 8 m when the AMR is in the vicinity of the access point and 22 m when the AMR navigates the northern part of the lab. Notably, there is line-of-sight (LoS) between the UE and the access point in most locations of the hall, except in the western area, where a rack with heavy machinery blocks the link. Fig. 7(b) shows the RSRP acquired on the navigated route. Note that this value is calculated as the average received signal in the synchronization signal blocks that the network transmits with 20 ms periodicity. Regarding the south location of the hall, the RSRP reaches values of around -60 dBm due to the LoS condition, and the short range of the link. In the northern part, these values decrease to values below -70 dBm due to the larger link range. It is notable that in the western part, there is an abrupt transition where the signal drops to -80 dBm due to the blocking of the signal given the machinery and racks present in the scenario.

To analyze in more detail the behavior of the RSRP over distance, Fig. 8(a) shows the relation between both parameters, where a decrease of the RSRP is observed as the distance increases due to propagation attenuation. To quantitatively evaluate the decrease in RSRP, the path gain can be modeled simply through a slope-intercept model which accounts for the path loss exponent n [34]. Considering the linear relation between RSRP and path gain given the fixed transmit power and the omnidirectional gain of the antennas in UE and AP, RSRP curves in terms of distance can be established for different values of path loss exponent n . Therefore, radio propagation conditions can be determined taking into account a loss exponent $n = 2$ for free space as a baseline. Referring to Fig. 8(a), it can be observed that the scatter plot tends to converge to values close to the curves denoted by $n = 3$ and $n = 4$. These values are similar to those observed in industrial environments with high machinery density in the state of the art [22], [35]. Finally, it is worth noting the region located within the range of 10 to 13 m, where a sharp drop in RSRP is observed. This region corresponds to the previously described

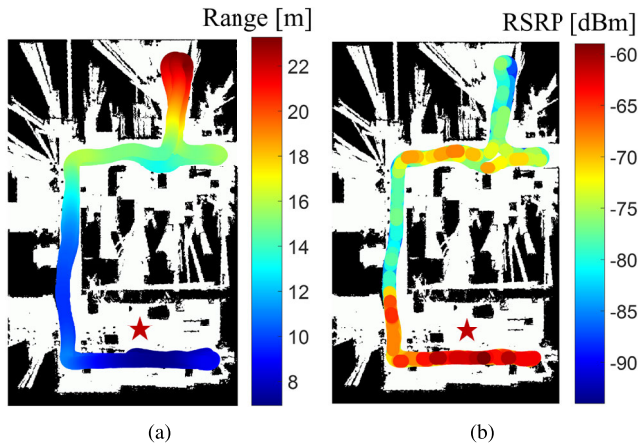


FIGURE 7. AMR navigated journey in the 5G SmartLab illustrating: (a) the range between the UE and access point, and (b) the RSRP in the link budget.

area that experiences total signal blockage, relying on signal reflections and diffractions in the environment rather than on LoS.

2) FORESTRY

Fig. 8(b) shows the RSRP values against the varying distance inside the deep forest. The RSRP measurements in a deep forest environment demonstrate a clear attenuation of signal strength with increasing distance from the NoW location. At a short distance of 15 m, the RSRP is strong at -35 dBm, indicating minimal signal loss. However, as the distance increases, the signal strength diminishes significantly. At 133 m, the RSRP drops to -68 dBm, highlighting the impact of dense foliage on signal propagation. Beyond 200 m, the signal degradation becomes more pronounced, with RSRP values reaching -87 dBm at 207 m and -98 dBm at 286 m. These measurements underline the challenging nature of maintaining robust wireless communication links in heavily forested environments, where signal absorption and scattering by vegetation contribute to substantial path loss.

3) LIVESTOCK TRANSPORTATION

Fig. 8(c) shows the RSRP over distance, measured at the farm location, and illustrates that connectivity can be maintained for distances up to about 60 m from the receiver, where the RSRP is about -115 dBm. Note that the RSRP measurements are independent of the bandwidth and the TDD configuration and thus all four curves in Fig. 8(c) have similar performance. Moreover, because the measurements were performed in LoS and there were not many obstacles in the environment, a close to free space propagation has been observed. The rather limited coverage achieved with the Amarisoft CallBox is due to its limited hardware capabilities, i.e., four simple and low-cost omnidirectional antennas and low transmit power. This showcases a clear trade-off between performance and investment in equipment.

4) MINES

The open-pit mine private network aims to cover different coverage challenges, including hotspots, enabling high data rates and capacity while ensuring a seamless user experience for heavy dump truck and excavator operators. The RSRP coverage measurements that are collected from the open-pit mine private network are shown in Fig. 8(d).

The measurement points are chosen based on the open-pit mine area. We chose five points that model the performance of the heavy dump trucks and excavators. Five points are selected, four in the corner of the open-pit mine and one in the middle. These points belong to road side units that are based on specific locations rather than on a specific trajectory and they are connected to gNodeB 1 or gNodeB 2 (see Fig. 6 for their locations). Apart from the RSRP measurements shown in Fig. 8(d), the signal to interference and noise ratio (SINR) is also measured at the same locations as the RSRP, as shown in Fig. 9.

The results show that the RSRP and SINR do not decrease with distance, which happens due to two main reasons. The first reason is that road side units get service from two different base station towers (gNodeB 1 or gNodeB2) and there are available blockages during the performance evaluation. The second reason is that the base station towers are not designed and optimized specifically for the open-pit mine. These two base stations are commercially available by the mobile network operator and give service the surrounding villages, and private networks are deployed on them. The base station tower's antenna direction, tilt values, and height are chosen based on the surrounding village and road, not the open-pit mine. Road side units receive signal from the closest base station, and the base station sector does not directly point to the road side units, excavators or trucks.

B. UPLINK THROUGHPUT

Uplink throughput is a relevant KPI for all network deployments as all use cases have requirements on the quality of the uplink channel. For example, sufficient uplink throughput is required for uploading a video to a remote server for processing. Uplink throughput depends among others on the distance, transmit power, antenna gains, propagation environment, bandwidth, as well as the TDD configuration. For the different network deployments, the uplink throughput is shown for different distances.

1) INDOOR ENVIRONMENTS

To evaluate the AMR uplink performance in a realistic indoor 5G environment, tests were carried out at the 5G Smart Production Lab. During these experiments, the AMR navigated autonomously through the lab while an *iperf* client, running on the modem acting as the UE, measured the maximum uplink packet transmission rate using the UDP protocol. Throughput was recorded at one-second intervals along the robot's trajectory to capture variations in performance across the scenario. Fig. 10(a) shows the

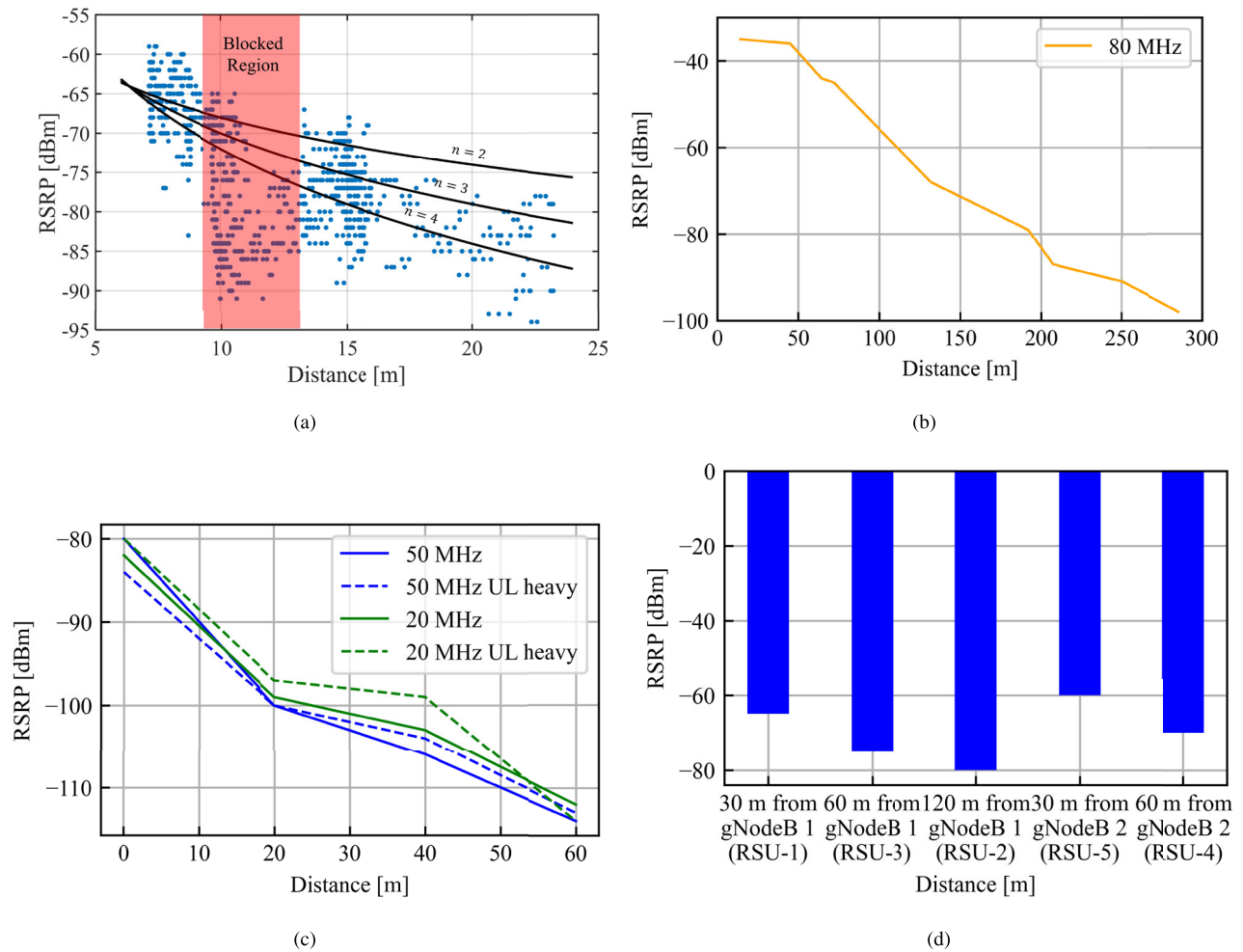


FIGURE 8. RSRP over distance (a) in the 5G SmartLab indoor scenario (path loss exponent curves $n = 2, 3$ and 4 are shown for reference), (b) in the deep forest, (c) at the farm, and (d) at the open-pit mine.

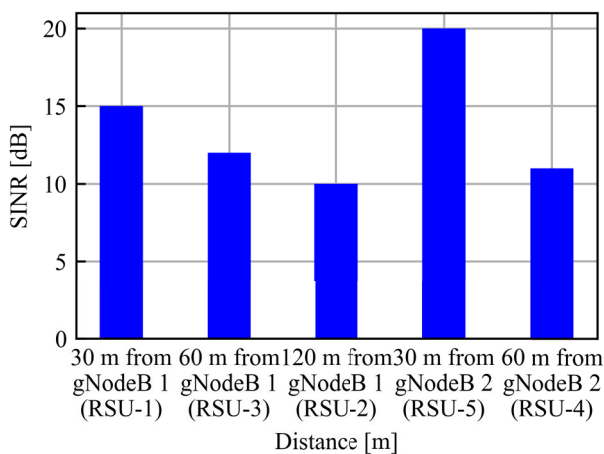


FIGURE 9. SINR over distance at the open-pit mine.

uplink throughput at different positions of the 5G smartlab given 100 MHz bandwidth and the TDD configuration of 3 uplink and 7 downlink slots. At short range, the network

is able to provide uplink throughput between 60-70 Mbps. However, when the AMR enters the NLoS condition or the range of the measurements increases, it is observed that the uplink throughput drops significantly to values below 30 Mbps. This fact can be explained from the link budget perspective, since the transmission power at the UE side might be constrained, not reaching RSRP or SINR values to establish modulation and coding schemes leading to large uplink throughput.

2) FORESTRY

Fig. 10(b) shows the uplink throughput against the distance which also corresponds to different RSRP values. The peak uplink throughput was achieved at the location with the maximum RSRP at a nearby distance of approximately 15 m whereas around 10 Mbps is achieved at a distance of 300 m deep in the forest. Moreover, the uplink throughput (and the coverage distance) can be further increased by increasing the transmit antenna height, the height of the receiving unit with multiple antennas and other features like beamforming.

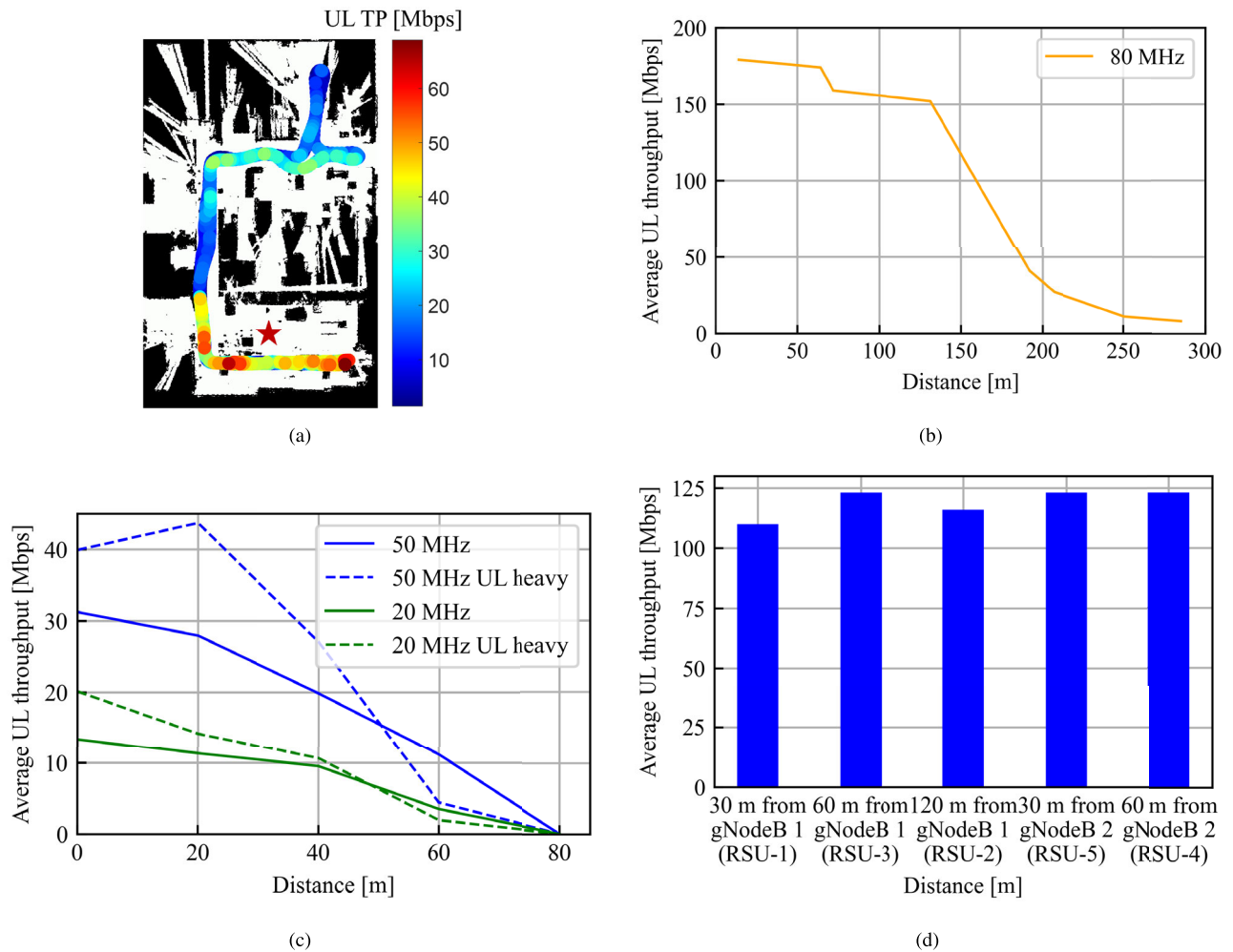


FIGURE 10. Average uplink throughput over different distances (a) in the 5G SmartLab, (b) in the deep forest, (c) at the farm, and (d) at the open-pit mine.

3) LIVESTOCK TRANSPORTATION

Fig. 10(c) shows the average uplink throughput versus distance for the considered bandwidth sizes and TDD configurations. Overall, it is illustrated that the larger bandwidth sizes provide higher uplink throughput, because generally throughput scales with bandwidth size. Additionally, Fig. 10(c) shows that the uplink heavy TDD configurations provide higher uplink throughput, which results from having more uplink slots available within the TDD frame. However, for distances more than about 50 m (or for RSRP lower than -100 dBm), the uplink heavy TDD configurations have a similar performance as their respective “regular” TDD configurations.

Fig. 10(c) shows that the measured average uplink throughput does not significantly exceed the 40 Mbps, which is considered as a modest performance. The limited performance is due to the same reasons highlighted in Section IV-A3. Specifically, the limited receiver antenna gain and transmit power impact the pilot measurements performed at the UE, based on which the UE configures its uplink

transmission, e.g. transmit power and modulation and coding scheme. Regardless of the modest uplink throughput achieved with the Amarisoft CallBox, the use case requirement of 10 Mbps has been achieved for distances up to about 40 m from the receiver (or RSRP of -100 dBm) when 20 MHz are used. For the configuration with 50 MHz, the 10 Mbps requirement is achieved for distances up to about 55 m from the receiver (or RSRP of -107 dBm).

4) MINES

A mobile network operator provides services that require different uplink throughput at the same time. The proposed private network is designed to handle a mix of a wide range of traffic types. For example, traffic due to a file upload can exploit a grant intended for a latency-critical service. The starvation problem can be seen in the public networks. In contrast, from day one, the private network architecture design mitigates it for latency-critical services like autonomous trucks or excavator data.

The uplink throughput measurements that are collected from the open-pit mine private network are illustrated in Fig. 10(d). The uplink throughput measured values are distributed between 100 and 125 Mbps in the measurement points. The different values have been monitored in both base stations. For the uplink, the available transmission power is significantly lower than for the downlink. Uplink throughput depends on the availability of the UE in the gNodeB. The measured values are not correlated with the RSRP, SINR and distance for the uplink.

C. DOWNLINK THROUGHPUT

Downlink throughput is a relevant KPI particularly for the use cases of indoor environments and forestry. For example, sufficient downlink throughput is required to support virtual reality glasses in education centers and remote assistance in the forest. Downlink throughput depends among others on the distance, transmit power, antenna gains, propagation environment, bandwidth, as well as the TDD configuration. For the two relevant network deployments, the downlink throughput is shown for different distances.

1) INDOOR ENVIRONMENTS

Following a procedure similar to the one presented in Section IV.B.1, Fig. 11(a) shows the downlink throughput measured with *iperf* throughout the scenario. Given the favorable resource allocation in the downlink due to the TDD configuration, the downlink throughput obtained is significantly higher than that uplink throughput seen in Fig. 10(a), with values up to 700 Mbps. These values decrease as the distance increases. However, in the worst case, a throughput of at least 100 Mbps is ensured for distances up to 22 m. It is noteworthy that when moving from LoS to NLoS condition, the throughput is reduced from 700 Mbps to around 400 Mbps, while for uplink, this value was reduced from 50 Mbps to 10 Mbps. Although in absolute terms the drop is larger, in relative terms the drop is lower, which can be explained by the fact that the access point has recovery mechanisms to avoid an abrupt SINR drop in the channel due to the larger flexibility in terms of power transmission. Fig. 12 shows the downlink throughput in terms of the different RSRP values acquired during the measurements. Both metrics show a positive correlation, with the locations whose RSRP is higher being those with the highest downlink throughputs in the scenario.

2) FORESTRY

Fig. 11(b) shows the achieved downlink throughput against distance in the forest. The peak throughput of around 650 Mbps is achieved around a distance of 120 m whereas the throughput of around 100 Mbps is achieved at a distance of around 300 m. The results show that signal attenuation is quite high and unpredictable in the forested environment due to the high vegetation and foliage environment.

D. LATENCY

Latency is a relevant KPI particularly for the use cases of indoor environments and forestry. For example, low latency is required to support remote surgery in healthcare centers and remote assistance in the forest. Generally, latency depends among others on the channel quality and achievable throughput, as well as the location of the user plane function, which will provide access to the Internet. For the two relevant network deployments, the latency is shown over time and distance.

1) INDOOR ENVIRONMENTS

One of the main advantages of 5G private networks is the reduced number of nodes that need to be routed to access the Internet, so that the end-user benefits from an ultra-reliable, low-latency system. To demonstrate this, Fig. 13(a) shows a round-trip time trace in the 5G private network for a predefined AMR path along the hall. This is done using the *ping* tool, which evaluates the round-trip time at 100 ms intervals. During the whole route, which accounts for approximately 7 minutes, the maximum round-trip time found is less than 25 ms, denoting the reliability of the network. By applying a moving average for 5 s intervals, it is observed that the average round-trip time value is in the range between 8 ms and 15 ms, taking into account that the route has mixed LoS and NLoS conditions. These results confirm the feasibility of carrying out critical low latency communications, thus enabling multiple use cases.

2) FORESTRY

Fig. 13(b) shows the latency and jitter values evaluated during the measurement campaign in the forest. The measured latency and jitter values for the 5G private network deployed in a forested environment exhibit notable variations across different distances, highlighting the challenges of maintaining consistent network performance in such conditions. Latency ranges from 12 ms to 27 ms, with fluctuations observed throughout the distance spectrum rather than a strictly increasing trend. For instance, higher latency values are recorded at both 45 m (26 ms) and 285 m (27 ms), while lower latency values, such as 12 ms, appear at intermediate distances (132 m). Similarly, jitter varies between 1 ms and 7.5 ms, with higher values observed at greater distances, such as 7.5 ms at 285 m. These results indicate that predicting latency and jitter in a forested environment is complex, as they are influenced not only by distance but also by factors such as foliage density, signal reflections, obstructions, and environmental conditions. However, despite these variations, the measured values remain well within acceptable limits, demonstrating that the 5G private network performs reliably even in challenging terrain.

E. BACKHAULING

Throughput of the backhauling link is an important KPI for the connection of a standalone private network to the Internet.

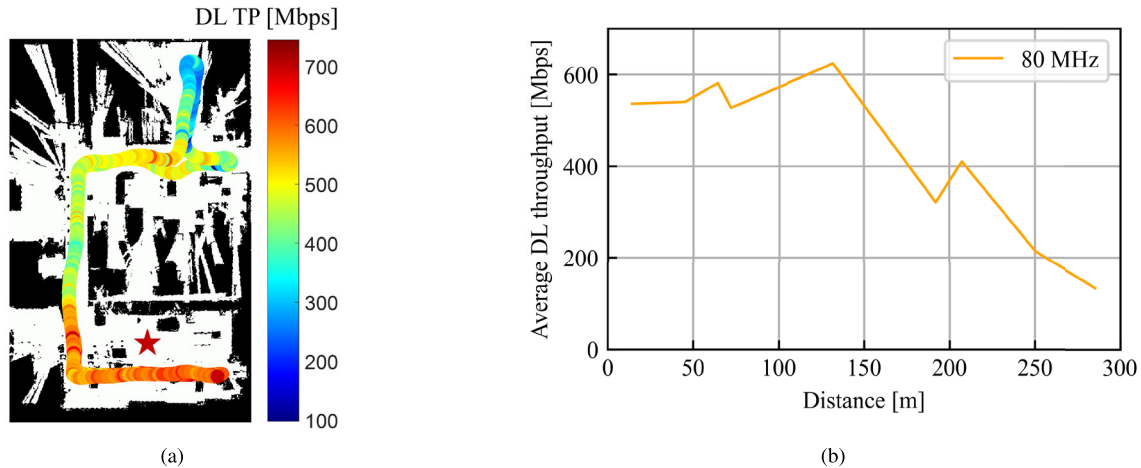


FIGURE 11. (a) Downlink throughput from the AMR navigated journey in the 5G SmartLab, and (b) average downlink throughput over distance in the forest.

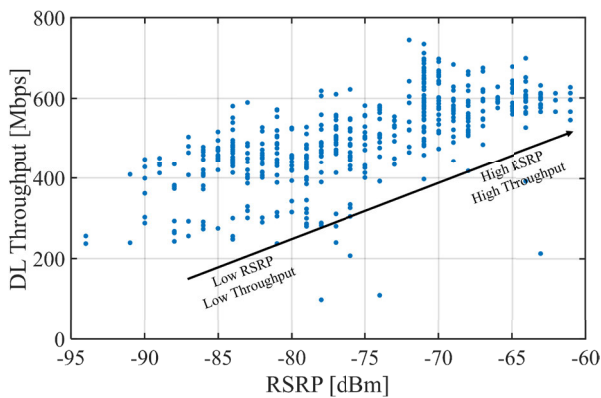


FIGURE 12. Downlink throughput over RSRP in the 5G SmartLab.

For the livestock transportation use case, it is assumed that no terrestrial infrastructure is available for backhauling and thus a satellite backhaul is required. The uplink throughput of the satellite-based backhauling link depends among others on the total traffic handled by the satellite, the location of the satellite and the weather conditions.

1) LIVESTOCK TRANSPORTATION

When satellite links are used for backhauling, it is likely that the end-to-end uplink throughput is dictated by the achievable uplink throughput of the satellite link. Therefore, we investigate the end-to-end uplink throughput between the UE and the iperf server, when the backhauling is provided by connecting the Starlink terminal to the Amarisoft. For the local 5G connectivity between the UE and the Amarisoft, the bandwidth was set at 50 MHz and the uplink friendly TDD configuration was applied, as this was the configuration providing the highest uplink throughput, as shown in Fig. 10(c).

Multiple measurements have been performed for the end-to-end uplink throughput revealing that the throughput ranges

between 9.41 Mbps and 21.3 Mbps. The observed variation of the uplink throughput is due to a.o. the position of the Starlink satellite, traffic from other users handled by the Starlink satellite, and the weather conditions. Moreover, the performance also depends on the applied Starlink subscription, which in this case was a roam subscription. Overall, the observed uplink throughput range is in line with the publicly reported Starlink performance in Europe [37].

Based on the performed measurements and the publicly reported Starlink performance in Europe, we conclude that the achievable uplink throughput is in the range of 10 Mbps to 20 Mbps. Comparing this range with the measured uplink throughput provided locally by the 5G private network, as shown in Fig. 10(c), we conclude that the satellite link is a limiting factor and acts as an upper-bound to the achievable uplink throughput. However, its performance is sufficient to support the livestock transportation use cases, which require 10 Mbps in the uplink channel.

V. COST BENEFIT ANALYSIS

Based on the performance evaluation, we provide a cost-benefit analysis between public and private 5G networks, outlining several recommendations for researchers and stakeholders involved in deployment, assessing the feasibility of this type of network in rural contexts. For the cost benefit analysis, two 5G private network deployments are considered, namely SNPN and PNI-NPN with end-to-end network slicing. SNPNs share no network infrastructure with the MNO, whereas a network sliced realized PNI-NPN has all infrastructure and functionality with the MNO. The reason to choose these two extreme deployments is clarity and conciseness of the cost benefit analysis. Any other “hybrid” 5G private network deployments are situational dependent and rely on the business and technical conditions between the private entity and the public MNO. For the cost analysis, the following categories are taken into account:

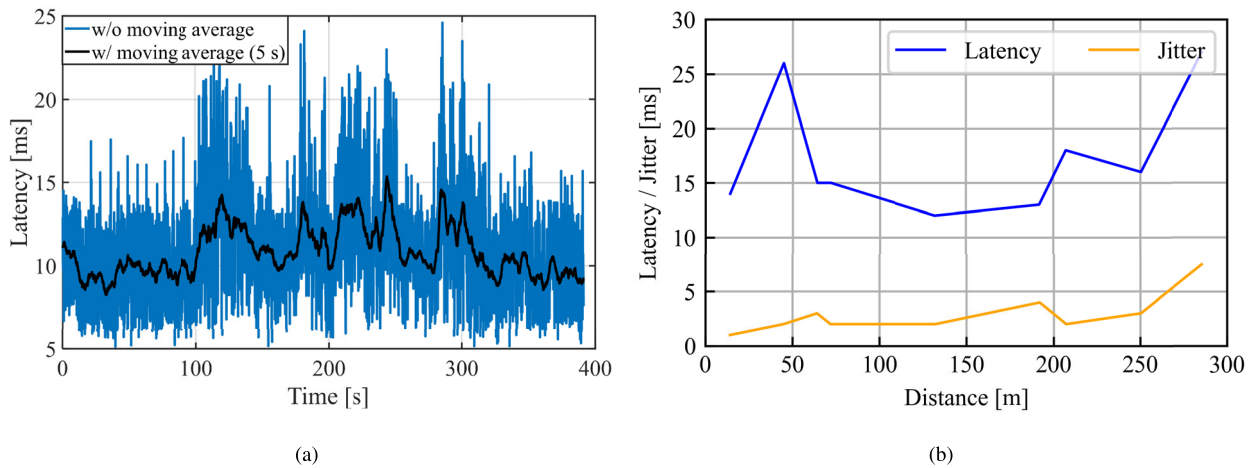


FIGURE 13. (a) Round-trip time temporal trace in the 5G Smartlab, and (b) latency and jitter over distance in the forest.

- *Spectrum costs:* For SNPNs this is the cost for using a local frequency license for operating the 5G private network. In Europe this is typically up to 100 MHz bandwidth in the spectrum range from 3.4 to 3.8 GHz (or in future 3.8 to 4.2 GHz). Typical costs are in the range of about 1000 Euro per year, or more depending on the coverage area of the 5G private network. In some European countries there are also exceptions of sub-usage of already allocated spectrum from MNOs (e.g. Finland, UK) [38], and in USA there is also a spectrum sharing framework via the Citizen Broadcast Radio Service (CBRS). For PNI-NPNs the usage of already allocated spectrum of the MNO can be assumed, which means low initial costs. However, the previous investments in spectrum by the MNO would be typically introduced in the subscription cost for the private entity.
- *Infrastructure costs:* For SNPNs this is the cost for installed infrastructure in terms of base stations for the wireless access part, all the transportation network needed, and the computing and storage hardware (e.g. servers) for deploying the network functions, core network, services platform and the operations and maintenance software. For PNI-NPNs the usage of already installed infrastructure by the MNO can be assumed. Note here that it should be expected that as MNO's infrastructure is shared also for the 5G private network purpose the associated infrastructure usage cost will be somehow included as fee in the operational and maintenance costs.
- *Operational and maintenance costs:* For SNPNs, this is the cost to keep the installed network up and running, monitor its functioning and service quality as well as upgrades of the installed infrastructure or network functionality. For PNI-NPNs existing operations and maintenance processes can be shared, and yet the operational costs can be increased by the MNO to account for the spectrum and infrastructure use.

For the benefit analysis, the following categories are taken into account:

- *Communication performance:* For SNPNs the throughput and delay of the communication links is more controllable/bounded due to customization (see also next category below) and typically low delays are achievable. As the results in this study show depending on the rural use case and 5G private network configuration download (upload) throughput of few hundreds (tens) of Mbps are achievable per user. From delay performance side field results show typical values of about 15-20 ms round trip time and below 5 ms jitter. For PNI-NPNs the throughput performance depends on the available coverage/capacity from the existing MNO infrastructure and there would be typically, higher delays.
- *Customization/scalability:* For SNPNs this is the benefit of customizing and scaling the network infrastructure for the desired performance and supported services closely tailored to the specific needs of the local enterprise. For PNI-NPNs there will be lower customization and scalability due to the shared MNO's infrastructure and inter-dependencies with other subscribers or private networks.
- *Security/Reliability:* For SNPNs this is the benefit regarding secure access and control of enterprise sensitive data and reliability of the connectivity service as the private entity is in full control of the communication data streams and the reliability of the installed infrastructure. For PNI-NPNs there is dependency on the security and reliability standards from the MNO. The eventual sensitive private entity data will be traversing and stored in public MNO's infrastructure.

Based on the above cost benefit analysis, the following deployment recommendations can be drawn.

Deployment of SNPNs is recommended if the public MNO infrastructure in the desired coverage area of the private entity is non-existent or providing poor coverage/capacity, and there

is somewhat larger operational private area requiring low delay connectivity. This could be applicable for example for large agricultural/mining enterprises that have sufficient budgets and require extensive coverage, high data throughput and low latency for applications like drone surveillance, autonomous machinery, and IoT sensors/actuators for highly critical business processes including high security and control for protecting sensitive enterprise data and ensuring reliable operations.

Deployment of PNI-NPN is recommended if the public MNO infrastructure can provide reasonable coverage/capacity in the desired coverage area of the private entity. Further, this deployment could be better suited for smaller-scale operations requiring moderate connectivity, including smaller private entities or those with budget constraints. Here, leveraging existing public MNO network infrastructure will reduce deployment and operational costs at the price of moderate connectivity services, such as basic IoT sensor networks for monitoring soil conditions, weather, and livestock and the private entity can afford sensitive data to be handled via public MNO infrastructure and can accept the reliability guarantees provided by the MNO.

VI. CONCLUSION AND FUTURE WORK

5G private networks have gained a lot of attention for deployments in industrial environments. Within the COMECT project, different 5G private network deployments have been proposed to address the connectivity challenges in remote areas and enable new use cases. In particular, use cases within four different environments have been considered, namely indoor environments (for e.g. healthcare or educational centers), forestry, livestock transportation, and mines. The proposed network deployments are diverse in terms of hardware (and thus cost) and configuration, and they have been evaluated via field trials. The results have shown that the deployed networks can support the presented use cases and that there is a clear trade-off between performance and cost. This highlights the need for designing 5G private networks based on the use case and the respective requirements, which could potentially lead to new business models and opportunities within the telecom sector. Therefore, this work has performed an empirical assessment and validation across challenging and underexplored remote scenarios, which are essential for understanding the feasibility and limitations of 5G private networks in rural contexts.

For future work, we propose to study the performance of other technologies which could potentially support the use cases, e.g. Wi-Fi. Then, a more in-depth cost benefit analysis will provide further insights on the applicability of 5G private networks, the opportunities that they bring and their challenges. Additionally, further analysis is recommended on the backhaul performance, as it can significantly limit the 5G private network performance. Apart from satellite communications for the backhaul, an alternative worth investigating is fixed wireless access. Finally, it is recommended to perform end-to-end measurements to investigate the impact of all the

components in the network and not only the performance of the 5G private network.

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